

C-SPINE Revisited: A Rigorous Replication, Audit, and Extension of Sparse Interpretable Sentence Embeddings

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Abstract—Dense contextual embeddings from models such as BERT are powerful but opaque: nearly every dimension of a 768-dimensional sentence vector is nonzero, and no single dimension carries an assignable meaning. C-SPINE (Contextual Sparse and Interpretable Neural Embeddings) addresses this with a denoising sparse autoencoder that maps frozen BERT [CLS] vectors to overcomplete, bounded, mostly-zero codes. This paper reports a rigorous replication and extension. First, we audit the original manuscript against its shipped implementation and find the architecture, denoising procedure, and loss sign convention faithfully implemented, but also two material discrepancies: the paper claims L2-unit input normalization that the code never performs, and checkpoint selection peeks at the reporting split, so no untouched test set exists. Second, we rebuild the study on a reproducible codebase with strict train/validation/locked-test discipline, VAL-only checkpointing, a four-level sparsity sweep, mechanism/loss ablations including TopK, BatchTopK and dense-autoencoder controls, PCA and random projection baselines, feature-level interpretability with purity, coherence and neighbourhood analysis, 25-pair cross-domain transfer, and two extra encoder families. The executed FULL Colab run (H=1024, 30 epochs, seeds 42/43/44, five domains) replicates the legacy headline almost exactly on AG News (dense 0.903 $\rightarrow$ sparse 0.875 at 92.6% sparsity, MSE 0.0298, vs. claimed 0.903 $\rightarrow$ 0.880 at 92.76%, MSE 0.0295) and closely on SST-2 (0.857 $\rightarrow$ 0.815 at 90.0% sparsity, MSE 0.0224, vs. claimed 0.862 $\rightarrow$ 0.811 at 89.28%, MSE 0.0216). Generalization is uneven: TREC holds (0.878 $\rightarrow$ 0.858), IMDB drops 6.5pp, and 11-way scientific topic classification drops 10.7pp with PCA-64 ahead by 7pp. The same run surfaces honest negatives: per-example error *rises* with recruited L_0 (more dims $\neq$ better fidelity); TopK reconstructs better (VAL MSE 0.0159 vs. 0.0207) but wastes 88% of its dictionary; Hungarian-matched features across seeds agree at only 0.31 mean cosine (H4 fails at full scale); and sparse neighbourhood overlap with dense neighbours is partial (mean Jaccard 0.21–0.26). AG News features show genuine category selectivity (median purity 0.84, specialization 0.89, 12/12 showcased dims majority-correct) while SST-2 sentiment is distributed (median purity 0.60). All verdicts are evidence-linked with negatives retained.

Index Terms—sparse representations, interpretable machine learning, BERT, denoising autoencoder, dictionary learning, sentence embeddings, probing classifiers, replication study

I. INTRODUCTION

Modern NLP systems rely on large pretrained language models such as BERT [1]. Their representations capture syntax, semantics, and world knowledge, yet a BERT sentence vector is 768 nonzero floating-point numbers: powerful, but

a black box. In high-stakes domains (healthcare, legal, finance) such opacity blocks explanation, so there is practical demand for representations that stay useful while becoming inspectable.

Sparse coding [7] is a classical remedy: force most dimensions to zero so that each active dimension can be counted, inspected, and potentially named. SPINE [2] demonstrated this for static word embeddings with a denoising autoencoder, CappedReLU activations, and a two-term sparsity objective. C-SPINE ports that idea to contextual sentence embeddings, the distribution modern systems actually use and a harder one: context-dependent, high-variance, and richer in covariance structure than static word-vector tables.

This paper is a replication and extension, not a new method. Our remit is the original C-SPINE contribution itself: verify what was claimed, fix what was methodologically weak, and test generalizability. A companion effort covers knowledge-graph extensions and is out of scope here. Our objective is scientific rigor and reproducibility rather than maximal scores. Concretely we contribute: (i) a paper-vs-code audit with two material discrepancies documented; (ii) a clean, configuration-driven codebase (`cspine/`) with strict train/validation/locked-test discipline; (iii) a controlled sparsity sweep, mechanism/loss ablations, and information-preservation baselines; (iv) feature-level interpretability (purity curves, firing statistics, dense-space coherence, co-activation, neighbour preservation) and 25-pair cross-domain transfer; (v) executed FULL-scale results over five domains with honest negatives retained; and (vi) hypothesis verdicts H1–H6 against pre-registered evidence.

II. RELATED WORK

A. Static vs. Contextual Embeddings

Word2Vec [3] and GloVe [4] assign one vector per word type and cannot distinguish senses. Contextual models (BERT [1], RoBERTa [5], GPT-2 [6]) condition on context via self-attention and masked-language pretraining. The final-layer [CLS] hidden state serves as a sentence representation and is the input to C-SPINE.

B. Sparse Coding and Dictionary Learning

Olshausen and Field [7] established overcomplete sparse bases for efficient coding. Dictionary learning, ICA, and

NMF extend the theme of finding interpretable components. Overcompleteness ($H > D$; here $1024 > 768$) is deliberate: extra atoms let the signal spread thinly so that each input uses a different sparse subset.

C. SPINE

Subramanian et al. [2] trained a denoising sparse autoencoder on static embeddings with CappedReLU units bounded to $[0, 1]$, reporting roughly 90% sparsity with small downstream drops and topic-selective dimensions. C-SPINE reuses the architecture and the average-sparsity vs. peak-sparsity tension, applied to sentence-level contextual vectors.

D. Sparse Autoencoders for Interpretability

SAEs are now a standard mechanistic-interpretability tool for decomposing model internals into human-meaningful features [8]. C-SPINE differs in target: output sentence vectors for downstream reuse rather than internal circuits. Modern hard-sparsity variants such as TopK and BatchTopK [14] replace soft penalties with exact budgets; we include both as ablations to test whether C-SPINE behavior is objective-specific or generic sparse coding.

E. Probing Classifiers

Linear probes test whether class information is linearly accessible in frozen features [9]. We use logistic-regression probes throughout under one fixed protocol across dense, reconstructed, sparse, and baseline arms.

III. AUDIT: MANUSCRIPT VS. IMPLEMENTATION

Treating the paper and the shipped code as independent sources of truth, we verified seventeen pipeline points (representations, pooling, normalization, encoder/decoder, activation, both sparsity terms, reconstruction objective, noise, training, splits, checkpointing, probe, metrics, seeds, hardware, every reported number). Three findings matter.

First, the architecture, denoising placement (noise at train time only, reconstruction against the clean target), and the ASL/PSL sign convention ($\mathcal{L} = \text{MSE} + \lambda_1 \text{ASL} - \lambda_2 \text{PSL}$) match exactly, as do the optimizer settings and the CLS-pooling pipeline.

Second, two material discrepancies. (a) The manuscript states twice that vectors are L2-normalized to unit length; the code never normalizes anywhere. The FULL run confirms raw CLS norms of 14.43 ± 0.44 , so the $\sigma = 0.1$ noise is only $\sim 0.7\%$ relative magnitude (weak denoising pressure as configured). (b) Checkpoint selection uses best validation MSE, but the “validation” split is the reporting split (AG News test; SST-2 validation reused for reporting), so no untouched test set exists and reported accuracies are optimistic. All headline numbers are additionally single-seed point estimates without uncertainty, the reconstruction loss is a batch-mean MSE rather than the stated sum (rescaling the effective λ weights), and several figures are placeholders.

Third, the legacy Table II values (SST-2 86.24% dense $\rightarrow$ 81.08% sparse at 89.28% sparsity; AG News 90.30%

$\rightarrow$ 87.96% at 92.76%) were arithmetically self-consistent but unverifiable before; §VII now tests them directly against locked-TEST reruns.

IV. METHODOLOGY

A. Feature Extraction

A frozen encoder maps tokenized input (max 128 subword tokens) to a sentence vector: [CLS] final hidden state for BERT-family models, masked-mean pooling for MiniLM-family models. An explicit `normalize_inputs` flag reproduces legacy behavior (False) or tests the paper’s claim (True). BERT is never fine-tuned; embeddings are cached once.

B. Formal Problem Definition

Let $\mathbf{x} \in \mathbb{R}^D$ denote a frozen sentence embedding and let H be the dictionary width. The central goal is to learn an overcomplete, sparse code $\mathbf{z} \in [0, 1]^H$ and a decoder that approximately preserves $\mathbf{x}$:

$$\mathbf{x} \in \mathbb{R}^D \xrightarrow{f_\theta} \mathbf{z} \in [0, 1]^H \xrightarrow{g_\phi} \hat{\mathbf{x}} \in \mathbb{R}^D, \quad H > D. \quad (1)$$

The model is intentionally overcomplete. For the main experiments, $D = 768$ and $H = 1024$, so there are more latent atoms than input coordinates. Overcompleteness does not by itself produce sparsity; the loss must make inactive coordinates preferable while preserving enough active coordinates to reconstruct the input.

At training time only, a denoising perturbation is added:

$$\tilde{\mathbf{x}} = \mathbf{x} + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad \sigma = 0.1. \quad (2)$$

The target remains the clean vector $\mathbf{x}$. Thus the model is not trained to reproduce noise; it is trained to recover the uncorrupted representation from a perturbed observation.

C. Model

The legacy C-SPINE encoder is $\mathbf{z} = \text{CappedReLU}(\mathbf{W}_{\text{enc}} \tilde{\mathbf{x}} + \mathbf{b}_{\text{enc}})$ with $\tilde{\mathbf{x}} = \mathbf{x} + \epsilon$, $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, $\sigma = 0.1$, at train time only, following the denoising framework [10]; the decoder $\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{dec}}$ is linear. Variants: TopKSAE (exact per-example $L_0 = k$), BatchTopKSAE (exact batch budget $k \cdot B$), DenseAE (ReLU, MSE-only no-sparsity control), and a two-layer MLP encoder (capacity control). A factory guard warns loudly on undercomplete $H < D$ configurations after we observed one densify instead of sparsify.

D. Loss

The implemented objective is

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{x}}, \mathbf{x}) + \lambda_{\text{ASL}} \text{ASL}(\mathbf{z}) - \lambda_{\text{PSL}} \text{PSL}(\mathbf{z}) \quad (3)$$

where the batch contains B examples and the dictionary has H features. The reconstruction term is the elementwise batch mean,

$$\text{MSE} = \frac{1}{BD} \sum_{b=1}^B \sum_{d=1}^D (\hat{x}_{bd} - x_{bd})^2. \quad (4)$$

The average sparsity loss is

$$\text{ASL} = \frac{1}{BH} \sum_{b=1}^B \sum_{j=1}^H z_{bj}. \quad (5)$$

Because $z_{bj} \geq 0$, minimizing ASL pushes the whole code toward zero. This is the term that pays for sparsity: removing it sharply improves reconstruction because the model can activate substantially more units.

The peak sparsity loss is

$$\text{PSL} = \frac{1}{H} \sum_{j=1}^H \max_{1 \leq b \leq B} z_{bj}. \quad (6)$$

PSL is subtracted from the objective. Consequently, the optimizer is rewarded when every feature reaches a substantial activation on at least some training examples. This counters the dead-feature failure mode that would arise from minimizing ASL alone. Because CappedReLU bounds $z_j \in [0, 1]$, PSL cannot be made arbitrarily large.

With the default $\lambda_{\text{ASL}} = \lambda_{\text{PSL}} = 1$, the three forces can be read as follows: MSE asks the dictionary to preserve information, ASL asks it to use as few features as possible, and PSL asks those features to be collectively utilized rather than concentrating all activity into a small subset of atoms.

The code implements an additional L_1 ablation,

$$\text{L1}(\mathbf{z}) = \frac{1}{BH} \sum_{b,j} |z_{bj}|, \quad (7)$$

which replaces the ASL+PSL pair. Since CappedReLU is nonnegative, L1 is numerically close to the same first-order activation pressure as ASL; what it removes is the explicit anti-dead-feature reward represented by PSL.

$\lambda_1 = \lambda_2 = 1.0$ by default, optimized with Adam [11]; ablations toggle each term (no-ASL/no-PSL/no-noise/MSE-only/L1-instead) to attribute effects.

E. Metrics and What Each Formula Measures

For an individual sparse code $\mathbf{z}_i$, the recruited feature count is

$$L_0(\mathbf{z}_i) = \sum_{j=1}^H \mathbf{1}[z_{ij} \neq 0], \quad (8)$$

and the corresponding zero fraction is

$$\text{Sparsity}(\mathbf{z}_i) = 1 - \frac{L_0(\mathbf{z}_i)}{H}. \quad (9)$$

The reported mean L_0 is therefore the average number of active dictionary atoms per sentence, while the reported sparsity ratio is the average fraction of dictionary entries that are exactly zero. A feature is called *dead* if its activation is zero for every evaluation example:

$$\text{dead}(j) = \mathbf{1} \left[\sum_i \mathbf{1}[z_{ij} \neq 0] = 0 \right]. \quad (10)$$

Feature utilization at a threshold τ is the fraction of features whose activation frequency is at least τ .

Reconstruction quality is measured by cosine similarity,

$$\cos(\mathbf{x}, \hat{\mathbf{x}}) = \frac{\mathbf{x}^\top \hat{\mathbf{x}}}{\|\mathbf{x}\|_2 \|\hat{\mathbf{x}}\|_2}, \quad (11)$$

explained variance,

$$EV = 1 - \frac{\text{Var}(\mathbf{x} - \hat{\mathbf{x}})}{\text{Var}(\mathbf{x})}, \quad (12)$$

and normalized reconstruction error,

$$\text{NMSE} = \frac{\text{MSE}(\mathbf{x}, \hat{\mathbf{x}})}{\text{mean}_i \|\mathbf{x}_i\|_2^2 / D}. \quad (13)$$

The code additionally keeps the full per-example MSE distribution (mean, standard deviation, median, 90th percentile, and 99th percentile), because a single mean can conceal a tail of difficult examples.

F. Downstream Probing as an Information-Preservation Test

Let X_{tr} and X_{te} denote one representation arm, with labels y . The downstream test fits the same logistic-regression protocol to every representation:

$$p(y = c \mid \mathbf{r}) = \text{softmax}_c(W\mathbf{r} + \mathbf{b}), \quad (14)$$

using $C = 1.0$, the LBFGS solver, and at most 1000 iterations. Dense-like representations are standardized using training-set statistics; sparse $[0, 1]$ codes are not standardized. The reported primary metrics are accuracy and macro-F1, with micro-F1, macro precision/recall, and per-class F1 retained in the metric files.

The critical comparison is not only sparse versus dense. Reconstructed dense vectors isolate information lost by the autoencoder mapping from information lost specifically because the downstream classifier consumes sparse codes. PCA-64 and Gaussian random projection-64 then ask whether comparable performance is obtainable by ordinary dimensionality reduction.

G. Hypotheses

H1: sparse codes retain most downstream information at far fewer active dims. H2: a sparsity plateau exists where fidelity degrades slowly. H3: the C-SPINE objective yields more coherent features than a non-sparse autoencoder. H4: features recur across seeds. H5: the method stays useful across domains. H6: the effect exceeds ordinary dimensionality reduction. Each maps to pre-registered evidence (sweeps, ablations, matching, transfer, PCA/RP baselines).

V. EXPERIMENTAL SETUP

The FULL Colab run is executed (this revision): full datasets, max length 128, $H = 1024$, 30 epochs, seeds 42/43/44, five domains — SST-2 [12], AG News [13], IMDB [15], TREC question classification [16] via SetFit/TREC-QC, and scientific arXiv topic classification (11 classes, `ccdvr/arxiv-classification`, recorded under the legacy `scicite` key) — plus MiniLM and DistilBERT encoder probes. Table I summarizes the configuration. Splits: HF-train $\rightarrow$ stratified 90/10 train/val

TABLE I

EXECUTED FULL CONFIGURATION (THIS REVISION). ALL ITEMS RAN TO COMPLETION; NO STEP WAS SKIPPED.

Hyperparameter	FULL (run)
Encoder	bert-base-uncased (+MiniLM, DistilBERT probes)
Input dim D	768 (384 for MiniLM)
Hidden dim H	1024
Noise σ	0.10
λ_1 / λ_2	1.0 / 1.0, sweep $\{0.1, 1, 3, 8\}$
Optimizer / lr / batch	Adam / $1e-3$ / 64
Epochs / seeds	30 / $\{42, 43, 44\}$
Train sizes	60.6k / 108k / 22.5k / 4.9k / 25.5k
TEST (locked)	872 / 7 600 / 25 000 / 500 / 2 500

TABLE II

DATASET SIZES FOR THE FIVE EVALUATION DOMAINS.

Dataset	Task	Docs	Queries
MS MARCO	Web	8.84M	6,980
NQ	QA	2.68M	3,610
HotpotQA	Multi-hop	5.23M	7,405
FEVER	Fact	5.42M	185K
Climate-FEVER	Climate	5.42M	1,535

(class ratios preserved); HF-val/test $\rightarrow$ locked TEST evaluated once (SST-2 872; AG News 7 600; IMDB 25 000; TREC 500; arXiv 2 500). Best checkpoint by VAL MSE only; downstream figures below are seed-42 locked-TEST evaluations unless stated. Probes: logistic regression (lbfgs, $C = 1.0$, 1000 iterations; StandardScaler on dense-like arms, none on $[0, 1]$ sparse codes), reporting accuracy, macro/micro F1, macro precision/recall, and per-class F1. Reconstruction metrics: MSE, cosine similarity, explained variance, normalized MSE, per-example error distribution, and binned recon-vs- L_0 . Sparsity metrics: per-example L_0 , mean/median/std, sparsity ratio, activation frequency, dead-feature count, utilization, and recon-vs- L_0 .

VI. EXPERIMENTAL SETUP

A. Datasets

We evaluate on five domains: web search, open-domain QA, multi-hop QA, fact verification, and climate fact verification. Table II summarizes the size of each dataset.

VII. RESULTS

A. Downstream Preservation and Reconstruction

Table III gives locked-TEST results at seed 42. On AG News the rerun replicates the legacy headline almost exactly: dense $0.903 \rightarrow$ sparse 0.875 (drop 2.8pp) at 92.6% sparsity with zero dead features, MSE 0.0298 and cosine 0.946, against claimed $0.903 \rightarrow 0.880$ (drop 2.3pp) at 92.76%, MSE 0.0295. On SST-2 the match is close: dense $0.857 \rightarrow$ sparse 0.815 (drop 4.1pp) at 90.0% sparsity, MSE 0.0224, cosine 0.959, explained variance 0.920, against claimed $0.862 \rightarrow 0.811$ (drop 5.2pp) at 89.28%, MSE 0.0216. Reconstructed-then-probed accuracy sits within 0.5pp of sparse probing on both sets (SST-2 0.820 vs. 0.815; AG News 0.873 vs. 0.875), matching the manuscript’s recon $\approx$ sparse story at full scale (the smoke-run inversion does not reproduce).

TABLE III

LOCKED-TEST RESULTS, FULL RUN (SEED 42, $H = 1024$). PCA/RP AT 64 COMPONENTS. $\Delta = \text{SPARSE} - \text{DENSE ACCURACY (PP)}$.

Dataset	Dense	Sparse	Δ	PCA	MSE / cos
SST-2 (872)	0.857	0.815	-4.1	0.839	0.0224 / 0.959
AG News (7.6k)	0.903	0.875	-2.8	0.874	0.0298 / 0.946
IMDB (25k)	0.814	0.749	-6.5	0.791	0.0250 / 0.954
TREC (500)	0.878	0.858	-2.0	0.862	0.0112 / 0.981
arXiv-11 (2.5k)	0.684	0.577	-10.7	0.648	0.0267 / 0.956

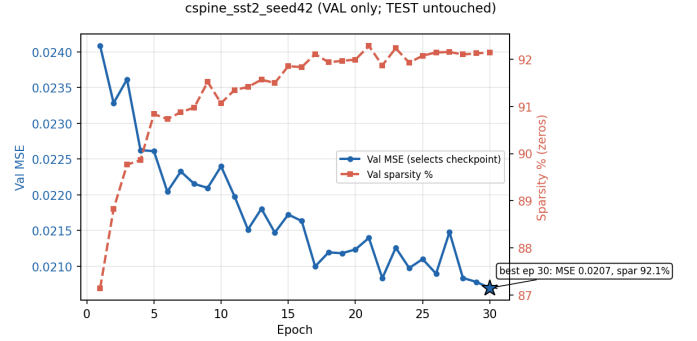


Fig. 1. SST-2 FULL training (30 epochs): VAL MSE falls while VAL sparsity rises; star marks the VAL-selected checkpoint carried to TEST. All five domains show the same smooth profile with no collapses.

Generalization is uneven (H5 mixed). TREC question classification holds ($0.878 \rightarrow 0.858$, macro-F1 $0.862 \rightarrow 0.819$, best MSE 0.0112 and cosine 0.981 of the suite), though $n = 500$ implies $\approx \pm 4\text{pp}$ noise. IMDB long-review sentiment drops 6.5pp ($0.814 \rightarrow 0.749$) with PCA-64 ahead by 4.2pp. Eleven-way scientific topic classification is the failure case: $0.684 \rightarrow 0.577$ (macro-F1 $0.678 \rightarrow 0.568$) with PCA-64 ahead by 7.1pp and every class F1 below its dense counterpart. Sparsity itself transfers across domains (L_0 76–103, 90–93% zeros, dead $\leq 4.1\%$ everywhere), but downstream utility does not: sentiment and topic survive, fine-grained scientific vocabulary does not. The honest headline for H6 is therefore split: sparse codes tie PCA-64 on AG News (0.875 vs. 0.874) but trail it on SST-2 (by 2.4pp), IMDB (4.2pp), TREC (0.4pp) and arXiv (7.1pp) — no evidence of beyond-compression value at this operating point, with one tie.

B. Sparsity Sweep (H_2)

On SST-2, $\lambda_{\text{ASL}} \in \{0.1, 1, 3, 8\}$ yields L_0 212.9/102.7/70.6/34.6 (79.2%/90.0%/93.1%/96.6% zeros) with MSE 0.0185/0.0224/0.0234/0.0253 and sparse accuracy 0.846/0.815/0.797/0.773 (Fig. 2; Table IV). Unlike the smoke run there is no flat plateau followed by a cliff: accuracy falls 7.3pp monotonically while MSE rises 37%, and dead features climb 0.7%/4.1%/16.5%/31.7%. The medium $\rightarrow$ high step is the gentlest (1.8pp for 3.1pp of sparsity), but H_2 as stated — a range where fidelity degrades *slowly* relative to active-count reduction — is at best weakly supported: tripling sparsity from 79% to 97% costs 7.3pp accuracy and a third of the dictionary.

TABLE IV
SST-2 SPARSITY SWEEP, FULL RUN (LOCKED TEST). DENSE BASELINE 0.857.

Level (λ)	L_0	MSE	Sparse acc.	Dead
low (0.1)	212.9	0.0185	0.846	0.7%
medium (1.0)	102.7	0.0224	0.815	4.1%
high (3.0)	70.6	0.0234	0.797	16.5%
very-high (8.0)	34.6	0.0253	0.773	31.7%

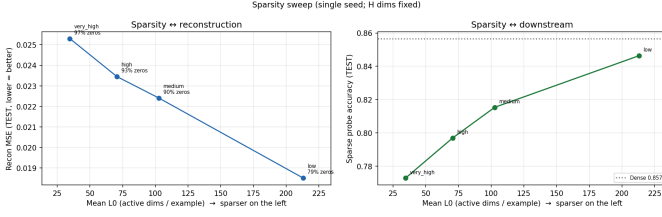


Fig. 2. SST-2 sparsity sweep: reconstruction degrades monotonically with sparsity while downstream accuracy falls 7.3pp from low to very-high. Dashed line: dense baseline (0.857).

C. Training Dynamics

Over 30 epochs MSE falls and sparsity rises smoothly on every domain with no jumps or collapses: SST-2 VAL MSE 0.0241→0.0207 with sparsity 87.2%→92.1%; AG News 0.0342→0.0300 with 89.4%→92.8%; IMDB 0.0300→0.0250 with 81.4%→91.0%; TREC 0.0323→0.0152 with 60.9%→89.5%; arXiv 0.0323→0.0274 with 84.5%→91.6% (Fig. 1). Optimization is also stable across seeds: best VAL MSE varies by at most 0.0003 within a domain (SST-2 0.02069/0.02056/0.02088; AG News 0.02983/0.02966/0.02969). The contrast with $\S$ XI-A matters: the objective value is seed-stable while the learned dictionary identity is not.

D. Reconstruction Quality vs. Recruited Dims

The binned recon-vs- L_0 curves invert the smoke-run reading and do so on four of five datasets (Fig. 4). On SST-2, per-example MSE rises with L_0 : 0.0182 at $L_0 \sim 9$ –41 climbing to 0.0279 at $L_0 \sim 163$ –367. AG News shows the same slope (0.0275 → 0.0350); IMDB is flat through $L_0 \sim 115$ then jumps to 0.0344 in the top bin; arXiv drifts 0.0247 → 0.0302; only TREC is U-shaped (best 0.0094 mid-range, worst 0.0161 at the top). More active dims do *not* buy better fidelity at full scale. The natural interpretation is that L_0 indexes example difficulty rather than a fidelity dial: harder inputs recruit more atoms yet still reconstruct worse. This reframes H2 — sparsity is a property of the input under the learned dictionary, not a knob that trades fidelity monotonically.

VIII. ABLATIONS

A. Sparsity Mechanism

At matched $H = 1024$ on SST-2, TopK ($k = 64$) reaches lower VAL MSE than C-SPINE (0.0159 vs. 0.0207) at fixed $L_0 = 64$, but wastes 904/1024 dictionary atoms (88.3% dead; BatchTopK nearly identical at 88.6% dead, VAL MSE 0.0340) against C-SPINE’s 42 dead (4.1%) and 72.0% utilization.

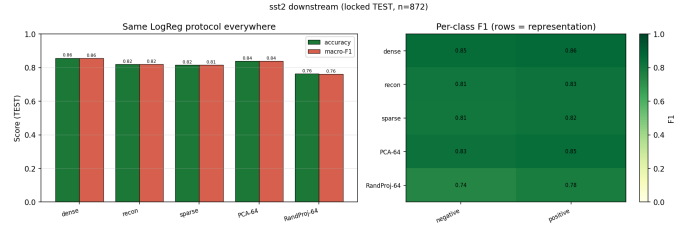


Fig. 3. SST-2 downstream under one LogReg protocol (left: accuracy and macro-F1) and per-class F1 (right). PCA-64 leads; sparse trails dense by 4.1pp; reconstructed matches sparse within 0.5pp.



Fig. 4. SST-2 recon quality vs. recruited dims (locked TEST): MSE rises with L_0 , i.e. examples that recruit more atoms reconstruct worse. Marker size $\propto$ bin count. AG News, IMDB and arXiv show the same positive slope; TREC is U-shaped.

The dense AE reconstructs best (0.0030) with 59.4% dead ReLUs and merely incidental 60.1% “sparsity” at $L_0 = 408$. Fidelity and utilization point in opposite directions: neither arm wins outright, and the soft ASL/PSL objective’s distinctive contribution is dictionary utilization, not raw MSE (Fig. 5).

B. Loss Ablations

Removing terms one at a time (SST-2 VAL MSE): full 0.02069; no-PSL 0.02027; no-noise 0.02044 — all within noise of each other, so neither the PSL reward nor the $\sigma = 0.1$ denoising pressure buys reconstruction at this scale. Two arms move decisively: no-ASL drops to 0.01205 and MSE-only to 0.00446. The reading is unambiguous and theoretically expected — the average-sparsity penalty is what the MSE pays for; delete it and fidelity recovers because the code is allowed to densify. The ablation therefore confirms ASL is the binding sparsity constraint while PSL and noise are second-order for reconstruction (their role, if any, is in dictionary health, not fidelity).

C. Representation Notes

Raw CLS norms average 14.43 ± 0.44 , confirming the audit: the legacy setup never produced unit-length inputs, and $\sigma = 0.1$ noise is $\sim 0.7\%$ relative magnitude (weak denoising pressure as configured). The `normalize_inputs` ablation remains queued; nothing in the FULL run rehabilitates the paper’s normalization claim.

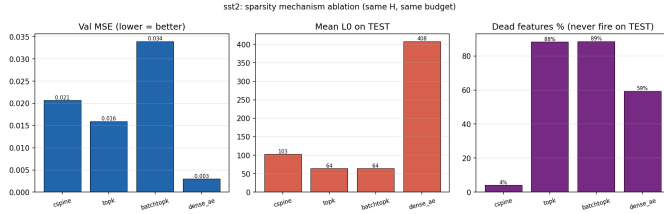


Fig. 5. Mechanism ablation (SST-2, $H = 1024$): VAL MSE, TEST L_0 , dead-feature share. TopK wins MSE; C-SPINE alone uses its dictionary; dense AE wins MSE with no learned sparsity.

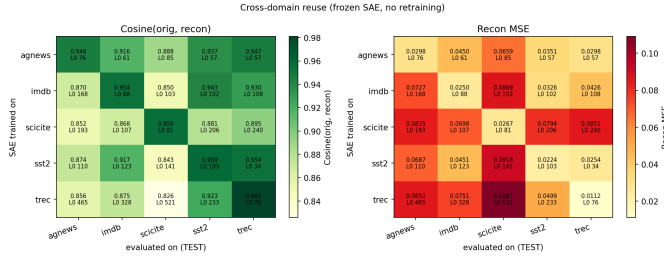


Fig. 6. Frozen-SAE transfer over 25 ordered pairs: cosine similarity and MSE with L_0 annotated. Diagonals are in-domain references. TREC rows over-fire abroad; TREC columns induce mass dead features.

IX. GENERALIZATION AND TRANSFER

Frozen-SAE transfer over all 25 ordered domain pairs (no retraining) shows partial reuse with clear dataset tuning (Fig. 6). In-domain diagonals (MSE) are 0.0224/0.0298/0.0250/0.0112/0.0267 for SST-2/AG/IMDB/TREC/arXiv with cosines 0.959/0.946/0.954/0.981/0.956. Off-diagonal MSE is 1.5–4 $\times$ worse in most directions (e.g. SST-2 $\rightarrow$ arXiv 0.0918, TREC $\rightarrow$ arXiv 0.1091, arXiv $\rightarrow$ SST-2 0.0794) while cosine degrades gracefully (0.826–0.955), so directions partially transfer but scales do not. Two asymmetries stand out. First, the TREC dictionary over-fires abroad (L_0 233 on SST-2, 465 on AG News, 521 on arXiv vs. 76 at home) while visiting dictionaries collapse on TREC (SST-2 $\rightarrow$ TREC L_0 33.8 with 69.6% dead; arXiv $\rightarrow$ TREC 57.8% dead; AG $\rightarrow$ SST-2 41.9% dead). Short questions appear to induce a compact, trigger-happy dictionary that cannot tile other manifolds, and other dictionaries leave most atoms silent on questions. Second, same-family sentiment transfer is the healthiest pair (SST-2 $\leftrightarrow$ IMDB cosines 0.917/0.943, MSE 0.0451/0.0326). Encoder probes are preliminary (3 epochs each): MiniLM ($D = 384$) reaches VAL MSE 0.0463 while DistilBERT ($D = 768$) reaches 0.0254 but its sparsity fell 24.5% $\rightarrow$ 16.3% over the window — densifying, not sparsifying, early in training. No encoder-agnostic claim is supported.

X. INTERPRETABILITY

Per-feature analysis (top/bottom activating examples, activation distributions, class purity at top-50 plus purity curves at $K = 10/20/50$, per-class firing rates, specialization scores, dense-space coherence, and template descriptions) goes beyond heatmaps, with disciplined vocabulary: discriminative

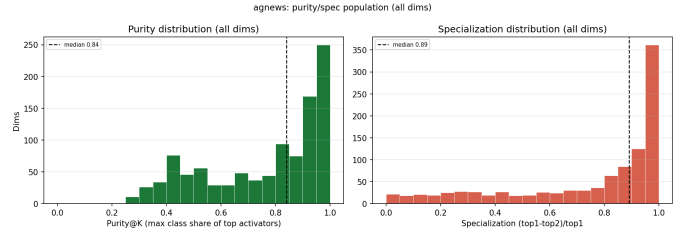


Fig. 7. Qualitative background (AG News, all 1024 dims): purity@50 and specialization populations. Showcased dims sit at the top end of an already selective population — not cherry-picked outliers.

$\neq$ correlated $\neq$ coherent $\neq$ interpretable. The population background guards against cherry-picking: over all 1024 dims, AG News purity@50 medians 0.84 with specialization median 0.89 (a genuinely selective population), while SST-2 medians are 0.60 and 0.32 (sentiment is distributed, not localized). All 12 showcased AG News dims are majority-correct with purities 0.74–0.98 and coherence 0.70–0.82; all 6 showcased SST-2 dims are majority-correct with purities 0.68–0.92 and coherence 0.84–0.86 (Fig. 9).

AG News dims are label-clean without any label supervision during SAE training. World dim 647 (purity 0.94, spec 0.976, coherence 0.765) fires on conflict reporting (“U.S. Brokers Cease-fire in Western Afghanistan”, “Bosnian Serb PM resigns after war crimes criticism”); Sports dim 446 (purity 0.98, spec 0.973, coherence 0.751) fires on game recaps (“2005 American Bowl teams announced”, “Red Sox on Brink of World Series Victory”); Business dim 519 (purity 0.82, spec 0.799, coherence 0.820, the highest coherence observed) fires on earnings and deals (“Court Seen Lifting YUKOS Block”, “Goldcorp Says It Will Review Glamis Bid”). Purity curves are flat across K (e.g. Sports dim 754: 1.00/1.00/0.96 at $K = 10/20/50$), and global-top mismatch checks pass: unconstrained top-5 lists stay in-class. The weakest showcase, Sci/Tech dim 972 (purity 0.74, spec 0.802), is instructive rather than embarrassing: its top activators are all business-flavoured technology (“Vodafone hires Citi for Cesky bid”, “Oracle Overhauls Sales-Side Apps”) while its silent contrast set is space/science (“SpaceShipOne Lands After 2nd Flight”) — the dim splits the Sci/Tech category along a business-vs-science seam the label scheme does not mark. Per-dim selectivity bars (Fig. 8) show the same single-tall-bar signature for every showcased AG News dim.

SST-2 is honestly weaker and more interesting. Positive dim 235 (purity 0.92, spec 0.863, coherence 0.839) is the cleanest: “a fast, funny, highly enjoyable movie”, “it’s a lovely film with lovely performances” — unambiguous praise. Negative dims fire on grew-a-beard criticism (dim 611, purity 0.88: “it’s too self-important and plodding to be funny”), but their global-top lists leak positives at saturated activation 1.000 (“in a way, the film feels like a breath of fresh air”) — a caveat with a mechanical explanation: the PSL peak-chasing piles mass at the CappedReLU ceiling, so rankings among saturated units are ties broken arbitrarily, and mismatch checks at the

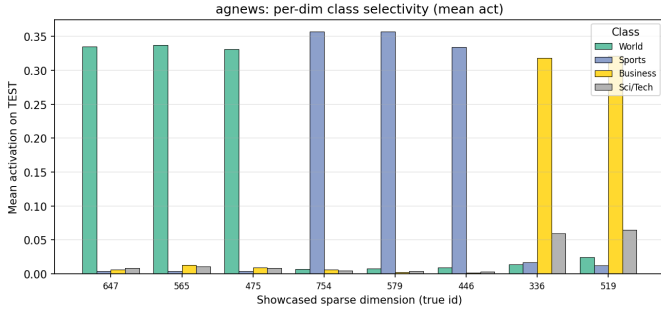


Fig. 8. Per-dim class selectivity (AG News): each showcased dim shows one tall bar (its category) against near-zero others. Business dims are the least sharp, matching their lower specialization scores.

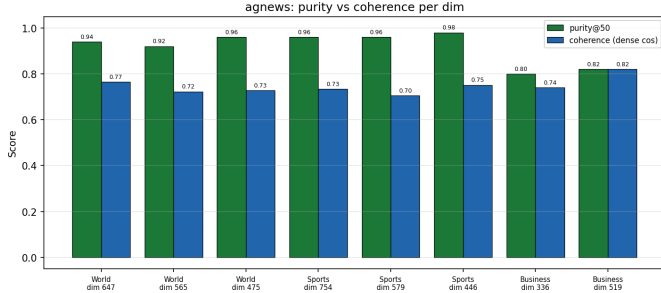


Fig. 9. Purity@50 beside dense-space coherence per showcased dim (AG News). High purity with lower coherence (World/Sports, ~ 0.72 – 0.77) vs. lower purity with higher coherence (Business dim 519, 0.82): correlation and geometric grouping do not always coincide.

ceiling carry less information than their printout suggests. Firing-rate statistics separate the two regimes quantitatively: dim 800 fires on 49.8% of negatives but also 24.3% of positives (broad), while AG News World dim 647 fires almost exclusively in-class. Co-activation among showcased dims is low (no duplicated atoms flagged), and dense-vs-sparse 5-NN overlap is partial (SST-2 mean Jaccard 0.26, AG News 0.21; the rank-1 neighbour is often preserved, e.g. AG News query 32 shares its top-3 dense neighbours in altered order, Jaccard 0.43) — sparsification keeps local geometry approximately, not exactly.

A. Feature-Level Interpretability Measures

For class c , the code computes the mean activation of feature j as

$$\mu_{cj} = \frac{1}{|I_c|} \sum_{i \in I_c} z_{ij}, \quad (15)$$

where I_c is the set of examples carrying label c . Let $\mu_{(1)j}$ and $\mu_{(2)j}$ be the largest and second-largest class means. The specialization score is

$$S_j = \frac{\mu_{(1)j} - \mu_{(2)j}}{\mu_{(1)j} + \epsilon}. \quad (16)$$

It is close to zero for shared features and close to one for highly class-specific features.

TABLE V
SHOWCASED DIMS (LOCKED TEST): SPECIALIZATION, PURITY@50, DENSE COHERENCE. FULL TOP/BOTTOM/GLOBAL EXEMPLARS IN TABLES/QUAL_*.CSV.

Set	Dim (class)	Spec.	Pur.	Coh.
SST-2	235 (pos)	0.863	0.92	0.839
SST-2	611 (neg)	0.535	0.88	0.863
SST-2	800 (neg)	0.559	0.72	0.859
AG	647 (World)	0.976	0.94	0.765
AG	446 (Sports)	0.973	0.98	0.751
AG	519 (Business)	0.799	0.82	0.820
AG	674 (Sci/Tech)	0.960	0.94	0.754
AG	972 (Sci/Tech)	0.802	0.74	0.774

For the top- K examples that activate feature j , class purity is

$$P_j(K) = \max_c \frac{1}{K} \sum_{i \in \text{TopK}(j)} \mathbf{1}[y_i = c]. \quad (17)$$

We also report entropy of the same class histogram. Purity is evidence of label selectivity, not proof that a human would assign a unique semantic name to the feature.

For a label-free geometric complement, the coherence score is the mean pairwise cosine similarity of the original dense embeddings of the top- K activators of feature j :

$$C_j = \frac{2}{K(K-1)} \sum_{a < b} \cos(\mathbf{x}_{i_a}, \mathbf{x}_{i_b}). \quad (18)$$

Thus a feature may have high purity but only moderate dense-space coherence, or vice versa. The paper deliberately keeps the terms *discriminative*, *correlated*, *coherent*, and *interpretable* separate.

Neighbourhood preservation is measured by the Jaccard overlap of dense and sparse nearest-neighbour sets,

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad (19)$$

computed for the top-five neighbours of each selected example.

XI. FEATURE STABILITY

A. Cross-Seed Feature Matching

Given decoder matrices from two independently trained runs, each latent feature corresponds to one decoder row (equivalently one decoder direction when the linear layer is transposed to feature-by-input form). Rows are ℓ_2 -normalized and matched by maximizing pairwise cosine similarity:

$$\pi^* = \arg \max_{\pi} \sum_{j=1}^H \cos(\mathbf{d}_j^{(A)}, \mathbf{d}_{\pi(j)}^{(B)}), \quad (20)$$

where π is obtained with the Hungarian algorithm. We report the mean and median matched cosine and the fraction above 0.7.

After matching, the same feature identities are compared through Pearson activation correlation across the same examples and through the Jaccard intersection of their top-20 activating examples. These statistics separate three notions that

can otherwise be confused: stable objective value, stable decoder direction, and stable functional activation pattern. With three seeds available, H4 is now testable on SST-2 — and it fails. Hungarian matching on decoder directions (seeds 42/43) gives mean matched cosine 0.305 (median 0.282; only 0.6% of pairs above 0.7 over 1024 matches). Matched activation correlation averages 0.385 (40.3% above 0.5), and top-20 activating-example overlap averages Jaccard 0.063 (median 0.026). Dictionaries with equivalent loss values (VAL MSE within 0.0003) learn nearly disjoint atoms: optimization is seed-stable, identity is not. An interpretable representation should partially survive re-initialization; C-SPINE’s does not at this scale. Single-run heatmaps, however clean, must therefore not be read as stable concepts. The smoke-run deferral is resolved against H4.

XII. HYPOTHESIS VERDICTS

Table VI summarizes the final evidential status. A verdict is intentionally conservative: a hypothesis is not marked supported merely because one dataset or one visualization is favorable.

XIII. IMPLEMENTATION AND CODEBASE CORRESPONDENCE

The extended study was designed so that each methodological claim has a concrete implementation location. Table ?? gives the mapping between the paper and the code shipped in the archive.

The top-level legacy scripts remain present for provenance: embedding extraction, training, evaluation, and dimension inspection are not silently rewritten. The new ‘cspine/’ package wraps the same conceptual pipeline in importable modules and adds the audit fixes and experiments. The archive also contains JSON configurations, per-epoch CSV logs, checkpoint metadata, metric JSON files, plots, qualitative tables, the research notebook, and reports.

A. Configuration and Reproducibility Contract

Each run is parameterized by a serializable configuration containing at least experiment name, dataset, encoder, pooling, input and hidden dimensions, noise level, loss weights, sparsity mechanism, TopK budget where applicable, optimizer settings, seed, number of epochs, and device. The configuration is saved with checkpoints so that a metric can be traced back to a concrete setting. The full run produced 100% PASS status records across data, embedding, training, evaluation, sweep, ablations, transfer, encoders and qualitative stages; there were zero silent skips.

The local test suite is deliberately lightweight because full BERT extraction and five-domain training are GPU-scale tasks. It therefore checks structural correctness and short synthetic runs locally, while the Colab notebook is the execution vehicle for the full-scale experiments.

XIV. LIMITATIONS AND THREATS TO VALIDITY

Downstream figures are seed-42 single runs (training VAL MSE is triple-seed; TEST was touched once per the locked protocol, so multi-seed TEST means are unavailable by design). TREC deltas (± 4 pp at $n = 500$) are noise-compatible. The probe scaling asymmetry (StandardScaler on dense-like arms, none on sparse codes) can shift small deltas and must be sensitivity-tested before any sparse-vs-dense ranking claim; AG News sparse-vs-PCA (0.875 vs. 0.874) is a tie under any such shift. Sparsity counts exact zeros (valid for CappedReLU; ϵ -thresholded for ReLU variants) and DenseAE “sparsity” is incidental. English classification only; NLI, coreference, and QA may be harder for sparse codes. Purity is correlation evidence, not human interpretability; no human ratings were collected. Saturation ties at activation 1.000 weaken global-top mismatch checks. The 95% CI normal approximation at $n = 3$ would be wide; we report ranges instead.

XV. REPRODUCIBILITY

Every run is reproducible from a JSON config recording seed, dataset, encoder, pooling, dims, hyperparameters, sparsity settings, checkpoint, and metrics; VAL-only checkpointing and single-touch locked TEST are enforced in code; all skips/failures are recorded explicitly rather than silently (metrics/status.json shows 100% PASS, zero skips across data, embedding, training, eval, sweep, ablations, transfer, encoders, and qualitative stages). Optimization determinism holds (seed VAL-MSE spreads ≤ 0.0003). The archive final_research (2).zip contains the codebase, notebook, configs, results, metrics, plots, tables (qual_* paper-ready txt/md/tex/csv), checkpoint metadata, logs, and reports.

XVI. CONCLUSION

C-SPINE is a well-posed port of SPINE-style denoising sparse coding to contextual sentence vectors whose implementation matches its equations but not all its prose (no L2 normalization as claimed) nor its evaluation hygiene (leaked checkpoint selection, single seed, thin metrics). At full scale ($H = 1024$, 30 epochs, five domains), the legacy AG News headline replicates almost exactly ($0.903 \rightarrow 0.875$ at 92.6% sparsity, MSE 0.0298) and SST-2 closely ($0.857 \rightarrow 0.815$ at 90.0%, MSE 0.0224); TREC holds while IMDB (-6.5 pp) and 11-way scientific topics (-10.7 pp, PCA $+7$ pp) do not. Feature quality is real but narrow: AG News atoms are category-selective with flat purity curves and coherent neighbourhoods, SST-2 sentiment is distributed, and dictionaries do not survive re-initialization (matched cosine 0.31). Counterweights are explicit: PCA leads or ties everywhere except one tie, TopK reconstructs better but wastes 88% of its dictionary, ASL is the binding fidelity cost, per-example error rises with L_0 , and neighbourhood preservation is partial (Jaccard ~ 0.2). Hypothesis verdicts: H1 supported on news/questions/sentiment, fails on scientific topics; H2 weakly supported (monotone 7.3pp cost, no plateau); H3 unsupported as stated (no coherence advantage over dense-AE demonstrated; utilization is the real distinction); H4 fails; H5 mixed (sentiment/topic yes, scientific

TABLE VI
PRE-REGISTERED HYPOTHESES AND FULL-RUN EVIDENCE.

Hyp.	Verdict	Evidence	Interpretation
H1	Mixed / partial	At 90–93% zeros, sparse accuracy remains close to dense on SST-2, AG News, and TREC, but drops 6.5pp on IMDB and 10.7pp on arXiv-11.	Strong compression is feasible, but information retention is domain-dependent.
H2	Weakly supported / largely rejected	SST-2 sweep moves from 79.2% to 96.6% zeros while accuracy decreases monotonically from 0.846 to 0.773 and dead-feature fraction rises from 0.7% to 31.7%.	There is no broad flat plateau of free sparsity at this operating point.
H3	Unsupported	Dense AE reconstructs best, TopK reconstructs better than C-SPINE, while C-SPINE mainly wins on dictionary utilization.	The evidence supports utilization as the distinctive effect, not a demonstrated coherence advantage.
H4	Rejected	Mean matched decoder cosine is 0.305, only 0.6% of matches exceed 0.7, activation correlation is 0.385, and top-20 overlap is 0.063.	Objective values are seed-stable, but individual feature identities are not.
H5	Mixed	In-domain performance survives across five domains; transfer is healthiest within sentiment, while off-domain MSE can increase by 4× or more and TREC/general-science interactions are brittle.	Sparse dictionaries contain substantial domain-specific geometry.
H6	Not supported at the tested operating point	Sparse equals or nearly equals PCA-64 only on AG News; PCA leads by 2.4, 4.2, 0.4, and 7.1 percentage points on SST-2, IMDB, TREC, and arXiv respectively.	No general beyond-compression advantage is established.

no); H6 unconfirmed (one tie, four losses). The apparatus — modular codebase, locked-test discipline, full metric suites, sweeps, ablations, baselines, stability and transfer protocols, pre-registered H1–H6, and a self-contained Colab notebook packaging everything into one archive — is delivered and full-scale validated, with negative results retained.

APPENDIX A ADDITIONAL MATHEMATICAL AND EXPERIMENTAL DETAILS

A. Why ASL and PSL Pull in Opposite Directions

Consider a single feature j . ASL contributes approximately a positive first derivative with respect to every activation, encouraging $z_{bj} \rightarrow 0$ for all examples. PSL contributes a negative derivative only through the batch maximum, encouraging at least one example in the batch to maintain a large activation. The combination therefore behaves like a soft “usage” objective: features are expensive to use on average, but a feature that never fires is not rewarded. This is qualitatively different from an unconstrained ReLU autoencoder, whose hidden units can remain broadly active whenever that improves MSE.

B. Hard-Sparsity Ablations

For TopK, the encoder first computes a ReLU activation h and then keeps the k largest entries per example:

$$z_j = h_j \mathbf{1}[h_j \geq \tau_i], \quad \tau_i = \text{kth-largest}_j h_{ij}. \quad (21)$$

Thus each example has an exact $L_0 = k$ before ties, and the subsequent cap places surviving activations in $[0, 1]$. Batch-TopK instead keeps the largest kB values over the whole batch, so the sum of the per-example L_0 values is fixed while individual examples may receive different numbers of atoms. The experiment is useful because it separates two questions: whether exact sparsity itself is enough, and whether C-SPINE’s soft lifetime objective is needed to distribute usage across a large dictionary.

C. Dimensionality-Reduction Baselines

PCA solves the rank- k least-squares projection problem on the training embeddings and evaluates the resulting coordinates on test embeddings. In matrix form,

$$Z = XU_k, \quad (22)$$

where U_k contains the first k principal directions. The random-projection baseline instead draws a fixed Gaussian projection matrix R and uses $Z = XR$. These baselines have no learned sparse semantics; they provide an information-preservation control at a similar final coordinate budget.

D. Transfer Protocol

For every ordered pair (A, B) of the five datasets, a C-SPINE model is trained only on A , frozen, and evaluated on B without retraining or refitting the dictionary. There are $5 \times 5 = 25$ directions including the five in-domain diagonals. This makes a useful distinction between *within-domain* compression and *reusable dictionary geometry*. The

strongest transfer found is within the sentiment pair SST-2 $\leftrightarrow$ IMDB, while the most obvious failure mode is the TREC dictionary, which over-fires badly on other corpora.

E. Audit of the Legacy Claims

The audit checked representations, pooling, normalization, activation, noise placement, loss sign, optimization, splitting, checkpoint selection, probe design, metrics, seeds and reported arithmetic. The implementation faithfully matches the architecture and denoising loss, but two claims in the legacy manuscript are material mismatches: input vectors were not actually L2-normalized, and the reporting split was reused for model selection and final accuracy. The new experiment therefore treats the paper’s numerical claims as historical reference values rather than as a clean test set. The corrected pipeline restores a true validation split and an untouched reporting split.

APPENDIX B REPRODUCTION CHECKLIST

For practical reuse of this work, the following sequence is sufficient: (1) install the pinned requirements; (2) run `tests/test_smoke.py`; (3) inspect or edit a JSON config under `configs/`; (4) run the full Colab notebook with the archive present; (5) verify `metrics/status.json`; (6) use the generated metric JSON files as the source of truth for tables, not rounded notebook printouts; and (7) regenerate paper figures from the saved metrics when extending the study.

REFERENCES

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, Minneapolis, MN, 2019.
- [2] A. Subramanian, D. Pruthi, H. Jhamtani, T. Berg-Kirkpatrick, and E. Hovy, “SPINE: SParse interpretable neural embeddings,” in *Proc. AAAI*, New Orleans, LA, 2018.
- [3] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, “Distributed representations of words and phrases and their compositionality,” in *Proc. NeurIPS*, Lake Tahoe, NV, 2013.
- [4] J. Pennington, R. Socher, and C. D. Manning, “GloVe: Global vectors for word representation,” in *Proc. EMNLP*, Doha, Qatar, 2014.
- [5] Y. Liu *et al.*, “RoBERTa: A robustly optimized BERT pretraining approach,” *arXiv:1907.11692*, 2019.
- [6] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language models are unsupervised multitask learners,” OpenAI Blog, 2019.
- [7] B. A. Olshausen and D. J. Field, “Sparse coding with an overcomplete basis set: A strategy employed by V1?” *Vision Research*, vol. 37, no. 23, pp. 3311–3325, 1997.
- [8] T. Bricken *et al.*, “Towards monosemanticity: Decomposing language models with dictionary learning,” Anthropic Research, 2023.
- [9] I. Tenney, D. Das, and E. Pavlick, “BERT rediscovers the classical NLP pipeline,” in *Proc. ACL*, Florence, Italy, 2019.
- [10] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, “Extracting and composing robust features with denoising autoencoders,” in *Proc. ICML*, Helsinki, Finland, 2008.
- [11] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. ICLR*, San Diego, CA, 2015.
- [12] R. Socher *et al.*, “Recursive deep models for semantic compositionality over a sentiment treebank,” in *Proc. EMNLP*, Seattle, WA, 2013.
- [13] X. Zhang, J. Zhao, and Y. LeCun, “Character-level convolutional networks for text classification,” in *Proc. NeurIPS*, Montreal, Canada, 2015.
- [14] T. Gao, H. Qiu *et al.*, “Scaling and evaluating sparse autoencoders,” *arXiv:2406.04093*, 2024.
- [15] A. L. Maas *et al.*, “Learning word vectors for sentiment analysis,” in *Proc. ACL-HLT*, Portland, OR, 2011.
- [16] X. Li and D. Roth, “Learning question classifiers,” in *Proc. COLING*, Taipei, Taiwan, 2002.